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# Module Code & Module Title

## CU6051NI Artificial Intelligence

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| | |
|---|---|
| **GitHub Link** | *https://github.com/PranavSharma195/Artificial_Intelligence* |

*I confirm that I understand my coursework needs to be submitted online via MST Classroom under the relevant module page before the deadline for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.*

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## 1. Introduction

Artificial Intelligence (AI) is a branch of computer science that enables computers and machines to stimulate human learning, problem solving, decision making and autonomy. AI has improved rapidly over recent decades due to significant improvement in data availability, computing power and researching (*What Is Artificial Intelligence (AI)? | IBM*, no date) . According to Standford AI Index Report, AI adoption has grown consistently across industries including finance, healthcare, transportation, and manufacturing, highlighting its increasing importance in solving complex real-world problems (AI Index | Stanford HAI, no date). Artificial Intelligence encompasses several subfields such as Machine Learning (ML), Natural Language Processing (NLP), Computer Vision, Robotics etc. Among these, Machine Learning is one of the most widely applied AI techniques because it enables systems to automatically learn patterns from historical data without being explicitly programmed. Mitchell defines machine learning as a system's ability to improve its performance on a task through experience (Mitchell, 2010). In this coursework, the concept of AI has been used for predicting price of used cars.

The used car market represents one of the largest segments of global automotive industry. Used cars market is driven by practical affordability, faster vehicle turnover and evolving consumer preferences. Unlike new cars which lose value significantly, used cars provide a cost-effective alternative for a wide range of buyers, including first-time owners, budget-constrained consumers, and those seeking specific models no longer in production. In economics terms, used cars are those assets to consumer whose value declines overtime with respect to age, usage, wear and tear and technological aspects. Research has documented that depreciation patterns are **non-linear**, often being steep in early years and flattening later, making accurate pricing models necessary for fair market valuation (Sodnomdavaa *et al.*, 2024). The used cars market is seen to provide benefit to developing countries where ownership of new car is less affordable. This leads to sustainable demand and a large secondary market system (Ma, 2024). Market data reflects on how much used cars have become economically significant. For example, the total annual worth of certain national used car markets can exceed tens of billions of U.S.

dollars, illustrating the scale and financial impact of this sector (Sharma and Kumar Mitra, 2024).

In summary, the used car market is large and evolving with high level of practical significance for everyday consumers. It also emphasizes theoretical knowledge for researchers. Its complexity arises from vehicle characteristics, market forces and consumer behaviour which makes it a viable domain for implementing machine learning techniques to improve price accuracy and market transparency.

## 1.1 Problem Domain

Predicting the price of used cars is a significant and well-studied problem in the field of artificial intelligence and data science. This topic falls at the intersection of market analytics, economic modelling and machine learning. The used car market is highly dynamic that many factors affect the price of the car which includes age, mileage and brand reputation. This complexity makes price estimation more challenging as a simple rule of thumb doesn’t work here. it requires sophisticated predictive modeling to capture non-linear relationships and interactions between features. Modern research has also shown that used car pricing benefits from machine learning techniques that can automatically learn complex patterns through historical data rather than relying on traditional methods alone. For example, a recent study evaluated that a combination of supervised and unsupervised learning approaches were used to predict the car prices. The results illustrated that advanced supervised methods such as XGBoost can effectively capture non-linear relationships between predictors and market prices, offering strong performance metrics such as high explained variance and low prediction error (Rao, Li and Wu, no date).

The existing body of research reflects several key characteristics and challenges which are inherent to problem domain and they are listed below:

### i. High Dimensionality and Feature Extraction

Used car datasets typically include many variables such as age, mileage, fuel type which ultimately affects the entire price of the used car. This makes it very hard to predict the price manually as each factor equally effects the price of the car. The figure below shows the relationship of different variables with car price and they are given below:

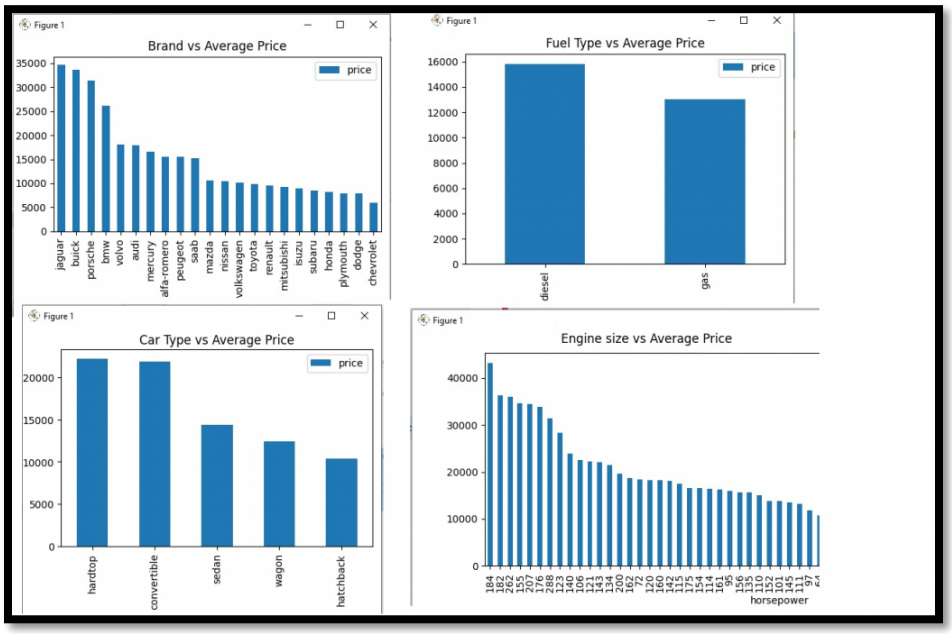


*Figure 1. Relationships among different variables*

### ii. Data quality and preprocessing

Realistic or real-world datasets often contain inconsistent and missing values which requires careful preprocessing and feature engineering. Techniques such as PCA (Principal Component Analysis) and clustering have been used to reduce dimensionality and improve model performance.

### iii. Choice of algorithms

Many studies shows that it is a very challenging task to choose the suitable algorithm for used car price prediction. So, comparison tables are made for determining the correct and the most accurate type of algorithms used for predicting this solution.

### iv. Forecasting

Used car price Used car price forecasting typically follows a trend pattern due to the depreciation of car value over time, along with seasonality that might arise from fluctuations in demand during certain months of the year (Khotimah, Swasono and Fajarianto, 2025). The following are the types of forecasting patterns that can be seen in the figure:

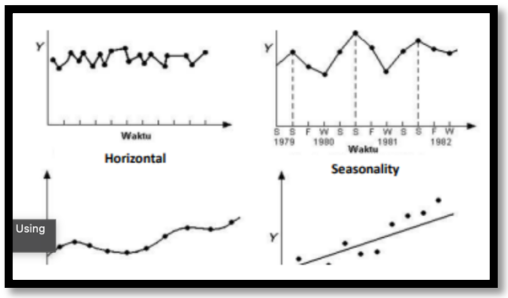


*Figure 2. Forecasting*

Overall, it is seen that used car price prediction is a complex predictive problem with direct applications in automotive sales, consumer decision-making, and business analytics. It also shows that effective solution requires deep understanding about the machine learning techniques but also careful feature engineering, data preprocessing and thoughtful selection of performance metrics.

## 1.2 Problem Statement and Objectives

### 1.2.1 Problem Statement

“The pricing of used cards is a very challenging task due to the influence of multiple factors such as vehicle age, mileage, fuel type, transmission, and ownership history. Traditional pricing methods often rely on manual calculations and subjective factors which lead to inaccurate and inconsistent price evaluation. Due to non-linear relationships among vehicle attributes, simple statistical approaches struggle to provide reliable predictions. With the growth in use of digital car marketplace, there is a need for automated and data driven solution. his project addresses the problem by applying supervised machine learning techniques to accurately predict used car prices based on historical data.”

### 1.2.2 Objectives

The objectives that are carried out through this project are listed below:

- To analyse key factors that influence car price such as age, mileage, fuel type, transmission etc.
- To preprocess and explore the used car dataset for meaningful patterns.
- To develop supervised machine learning regression models for price prediction.
- To compare the performance of different regression algorithms.
- To evaluate model accuracy using appropriate performance metrics.
- To identify the most effective model for predicting used car prices.
- To solve a real-world problem by making car price prediction completely automatic without needed any manual labour for calculation.

## 1.3 Dataset Used

The performance and reliability of a text or emotion classification system highly depend on the quality of the dataset used for training and testing the model. A high-quality dataset is a must for accurate results prediction. The dataset must properly represent the problem domain containing meaningful features and should be sufficiently large for learning patterns that generalize well to unseen data. Additionally, the source of dataset matters which means that trustworthy platforms provide well-structured files with consistent formatting and comprehensive metadata, which are crucial for reliable model development.

### 1.3.1 Dataset Characteristics

**i. Source**

The car price prediction dataset is publicly available in Hugging Face which is a popular website for datasets extraction, and it provides high quality and refined datasets for Machine Learning Tasks. The dataset is uploaded by the user VarunKumarGupta2003. Although the datasets Readme file is empty, its schema and tabular preview review its structure. The dataset has been downloaded and used by researchers and practitioners interested in supervised regression tasks. Hugging Face's infrastructure supports reliable data versioning and metadata, enabling reproducible experiments and integration with machine learning workflows.

To access the dataset, click "Car price dataset"

**ii. Size**

The dataset contains a single split labelled train with 10,000 rows of data, which is sufficient for model development. The dataset is provided in a csv format which makes it very easy to work with. Each row corresponds to one car record with complete information on features that potentially influence its price. The dataset size falls within

a typical range for regression modeling tasks in tabular machine learning. The size of dataset is 636 KB which makes it very fast to process and compute the results.

### iii. Features and Data Format

The datasets features are a mix of categorical (i.e. Make, Model, Fuel Type, Transmission) and numerical attributes (i.e. Year, Engine Size, Mileage). The target variable, Price, is stored as a floating-point value representing the car's resale value in a continuous range. Categorical values are represented as string classes, while numerical variables are stored in appropriate numeric formats. This structured format allows for straightforward preprocessing, including encoding categorical variables and scaling numeric features, which are crucial steps in supervised learning

### iv. Dataset Column Attributes

| Column Name | Data Type | Description | Possible Values |
|---|---|---|---|
| Make | String (Categorical) | Brand or company manufacture of the car. | 5 different brand names. |
| Model | String (Categorical) | Specific model variant of the car. | e.g. Model A, Model B |
| Year | Integer (Numerical) | Year of car manufacture. | Around 2000-2022 |
| Engine Size | Float (Numerical) | Engine displacement or capacity. | Continuous range roughly 1.0–4.5 L |
| Mileage | Integer (Numerical) | Total kilometres driven by the cart | Range from 5 to 200,000 kms |
| Fuel Type | String (Categorical) | Type of fuel used by the car. | Diesel, Petrol and Electric |
| Transmission | String (Categorical) | Type of gear system. | Manual or Automatic |
| Price | Float (Numerical-Target Variable) | Resale price of the car. | Values from 8,660–40,700. |

*Table 1. Dataset Column Attributes*

## 1.4 Type of Algorithm Used

This project employs Supervised Learning with Regression Techniques, which is a category of machine learning where models are trained using labelled data. In supervised learning, each input instance is associated with a known output label which allows the model to learn a direct mapping between the input and the target variable.

Supervised Learning is particularly suitable for this project as there are explicit input features and a corresponding target variable for the input features. Since, in this project the input parameters and the target variable (car price) is already known, supervised learning is used. This allows the model to be trained in such a way that it minimizes the prediction error by comparing predicted values with the actual prices. This makes supervised learning ideal for problems that require accurate and reliable predictions based on past data. Supervised learning is divided into classification and regression tasks. Classification tasks are used when the output variable is categorical or discrete. In this project, Regression is used because the output variable is continuous rather than categorical. Since the objective of this project is to predict the numerical price of a used car, regression techniques are the most appropriate choice.

In conclusion, this project uses supervised learning with regression techniques. This is chosen due to the availability of labelled data and the need to predict continuous output value. This approach enables accurate, scalable, and data-driven estimation of used car prices based on historical market data.

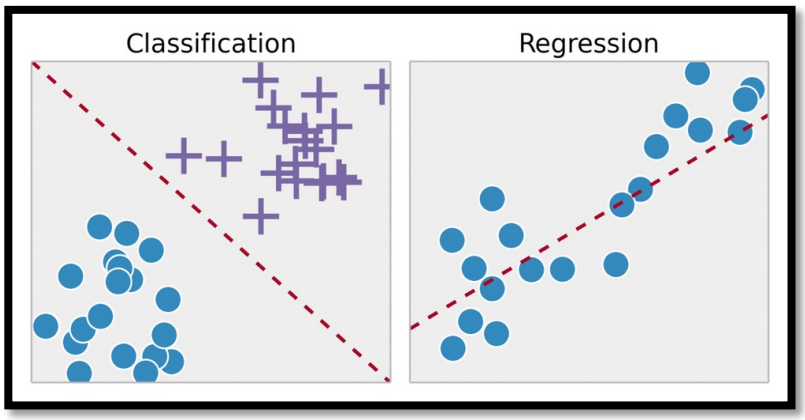


*Figure 3. Classification and Regression*

## 2. Background

In recent years, supervised learning-based regression models have become a reliable and widely accepted approach for predicting used cars prices by treating price as a continuous target variable and learning relationships between vehicle attributes and market value. Accurate used car price prediction provides valuable insights for multiple stakeholders, including buyers, sellers, dealerships, insurers, and financial institutions, by reducing pricing uncertainty and improving transparency in secondary automobile markets. As used car valuation depends on numerous interdependent factors such as age, mileage, brand, fuel type, transmission, and engine characteristics, traditional rule-based pricing methods often fail to capture the complexity of real-world pricing dynamics.

Research trends show gradual transition from a simple statistical models to advanced machine learning techniques to predict the price of used cars. Earlier studies relied on linear regression models to identify depreciation patterns. According to studies in applied machine learning for automotive analytics, algorithms such as Random Forest Regression, Gradient Boosting, and XGBoost consistently achieve lower prediction error and higher explanatory power compared to traditional regression methods (Pal, Arora, Kohli, et al., 2017).

### 2.1 Methods and Algorithms Used in Literature

Existing literature on used car price prediction frequently evaluates multiple supervised learning techniques. Linear models such that Multiple Linear Regression and Ridge Regression are commonly used for predicting the price of used cars. Tree-based regression models, including **Decision Tree Regression and Random Forest Regression**, are widely used due to their ability to capture non-linear patterns and handle mixed data types. Studies show that Random Forest models perform well in used car price prediction by reducing variance through ensemble averaging (Breiman, 2001) More advanced boosting techniques such as **Gradient Boosting and XGBoost** have been shown to further improve accuracy by sequentially minimizing prediction error and learning complex feature dependencies (Chen and Guestrin, 2016).

## 2.2 Dataset and Feature Engineering Practices

Recent studies show that the datasets required for model training was mostly extracted from marketplaces, dealership listings and public repositories which contains thousands of historical records with features such as brand name, model, odometer, fuel type etc. Several studies highlight that proper feature transformation significantly improves regression performance. For example, representing vehicle age instead of manufacturing year, log-transforming mileage, and separating categorical variables such as fuel type and transmission improve interpretability and predictive accuracy(Asghar et al., no date).

## 2.3 Related Research Papers Review

Below are 5 research papers that show work been done in same problem domain:

- **Study 1: Empirical Evaluation of Supervised Models (Rao, Li and Wu, no date)**

  This research evaluated multiple supervised learning algorithms including Linear Regression, Random Forest, SVR, Elastic Net, and XGBoost. The results demonstrated that XGBoost achieved the highest predictive accuracy and lowest error metrics, emphasizing the advantage of boosting techniques in handling heterogeneous used car data.

  **Dataset Used:** For this research, publicly available dataset which is readily available in Kaggle was used containing 10,000 records with features like car make, model, year, mileage, fuel type, and price.

- **Study 2: Machine Learning-Based Car Price Prediction (Li, 2024)**

  This study analysed the performance of Linear Regression, Decision Tree Regression, and Random Forest Regression. The results indicated that tree-based ensemble models consistently outperformed linear models in terms of $R^2$ and RMSE, particularly when the dataset included mixed numerical and categorical features.

  **Dataset Used:** For this research, a dataset of 8500 used cars collected from Chinese used car platforms and dealerships which included attributes such as year, mileage, engine size, brand, and price.-

- **Study 3: Linear vs Ensemble Models (Pal, Arora, Sundararaman, et al., 2017)**

This study compared traditional regression techniques such as linear regression with Random Forest Regression used for car price prediction. The results showed that Random Forest significantly outperformed linear regression by predicting more accurate results.

**Dataset Used:** Approximately 6,000 real-world used car listings collected from multiple online car marketplaces including CarDekho and public auction data. The dataset contained features such as year, mileage, fuel type, transmission, and engine size.

- **Study 4: Comparative analysis of Regression Algorithms (Jain and Parmar, 2022)**

  This study compared Linear Regression, Support Vector Regression, Decision Tree Regression, and Random Forest Regression. The results concluded that ensemble-based models provide more reliable predictions for used car prices due to their ability to model complex feature interactions and reduce overfitting.

  **Dataset Used:** For this research, used car dataset from Kaggle which contains 7500 rows was used with features including brand, year, mileage, engine size, fuel type, transmission, and price.

- **Study 5: Performance Analysis of Regression Models (R and C, 2023)**

  This research conducted a performance comparison of several regression algorithms and found that Random Forest and Gradient Boosting consistently outperformed simpler models. The study reinforced the effectiveness of ensemble learning for real-world used car price prediction tasks.

  **Dataset Used:** For this research, dataset of 6200 used cars scraped from online marketplaces including CarDekho and Cars24 with features including year, mileage, brand, model, transmission, fuel type, and price.

# 3. Proposed Solution

## 3.1 Algorithms Used

For predicting used car prices based on features like year, mileage, brand, fuel type, engine size, and transmission, the following three supervised regression algorithms will be used:

### i. Linear Regression

Linear Regression is a simple and widely used statistical method for predicting a continuous target variable based on one or more input features.

In this project, it models the relationship between car attributes like year, mileage, engine size, fuel type, transmission, brand and the car price.

The formula for Linear Regression is given below:

$$Y = B_0 + B_1X_1 + B_2X_2 + \cdots + B_nX_n$$

Where:

$Y$ = Predicted Price

$X_i$ = Features

$B_i$ = Coefficients

Linear Regression can further be understood with the help of the graph shown below:

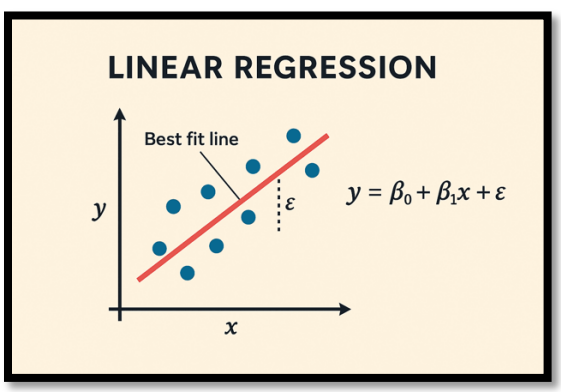


*Figure 4. Linear Regression*

### ii. Random Forest Regression

Random Forest is an ensemble learning method that combines multiple decision trees to make predictions. Each tree is trained on a random subset of the data and features, and the final prediction is the average of all trees. In this technique, each tree splits nodes based on criteria like **Mean Squared Error (MSE)** to minimize prediction error. It captures non-linear and complex relationships in the data.

Random Forest Regression is perfect for used cars datasets where there is a combination of categorical and numerical data. This algorithm can handle non-linear relationships such as the effect of mileage or car age on price. The workflow of Random Forest Regression can be further understood by the image shown below:

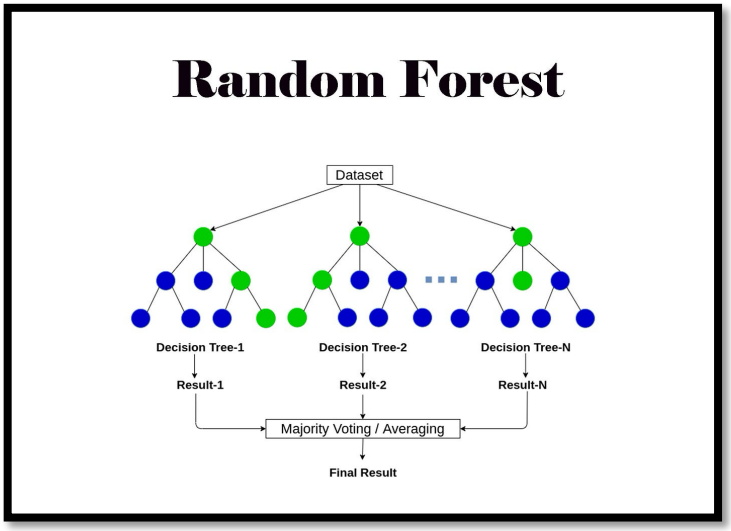


*Figure 5. Random Forest Regression*

### iii. XGBoost Regression

XGBoost is an optimized gradient boosting algorithm designed for speed and high performance. It builds trees sequentially, where each tree tries to correct the errors of the previous trees. It minimizes a loss function using gradient descent. This algorithm often outperforms other algorithms in tabular dataset. It includes regularization which improves generalization.

XGBoost is an excellent choice for this project because features interact in complex way. It often achieves higher accuracy and lower error metrics compared to linear models and traditional ensemble methods. This makes it suitable for this project of used car price prediction. The mathematics behind XGBoost can further be understood with the help of the figure shown below:

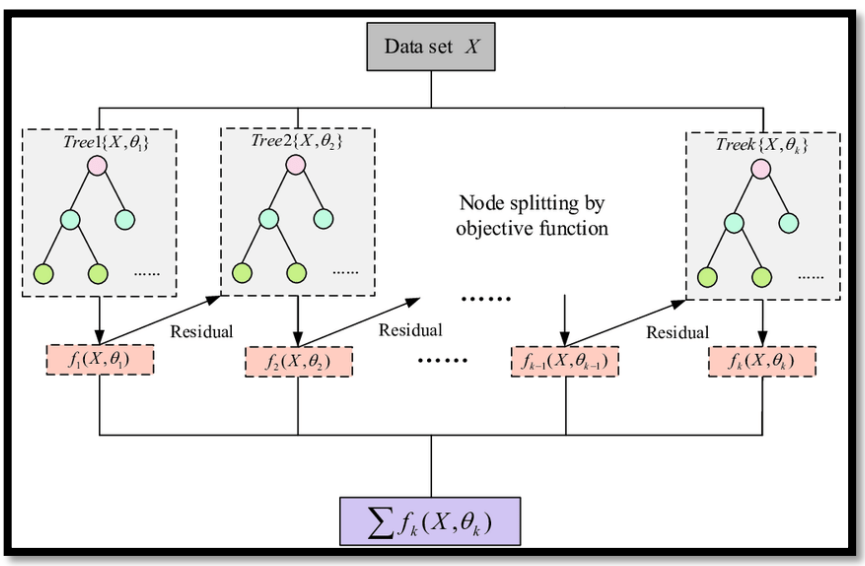


*Figure 6. XGBoost Regression*

### 3.1.1 Comparison among different algorithms

The image below visualizes a general idea about the algorithms being used to predict the price of second-hand cars:

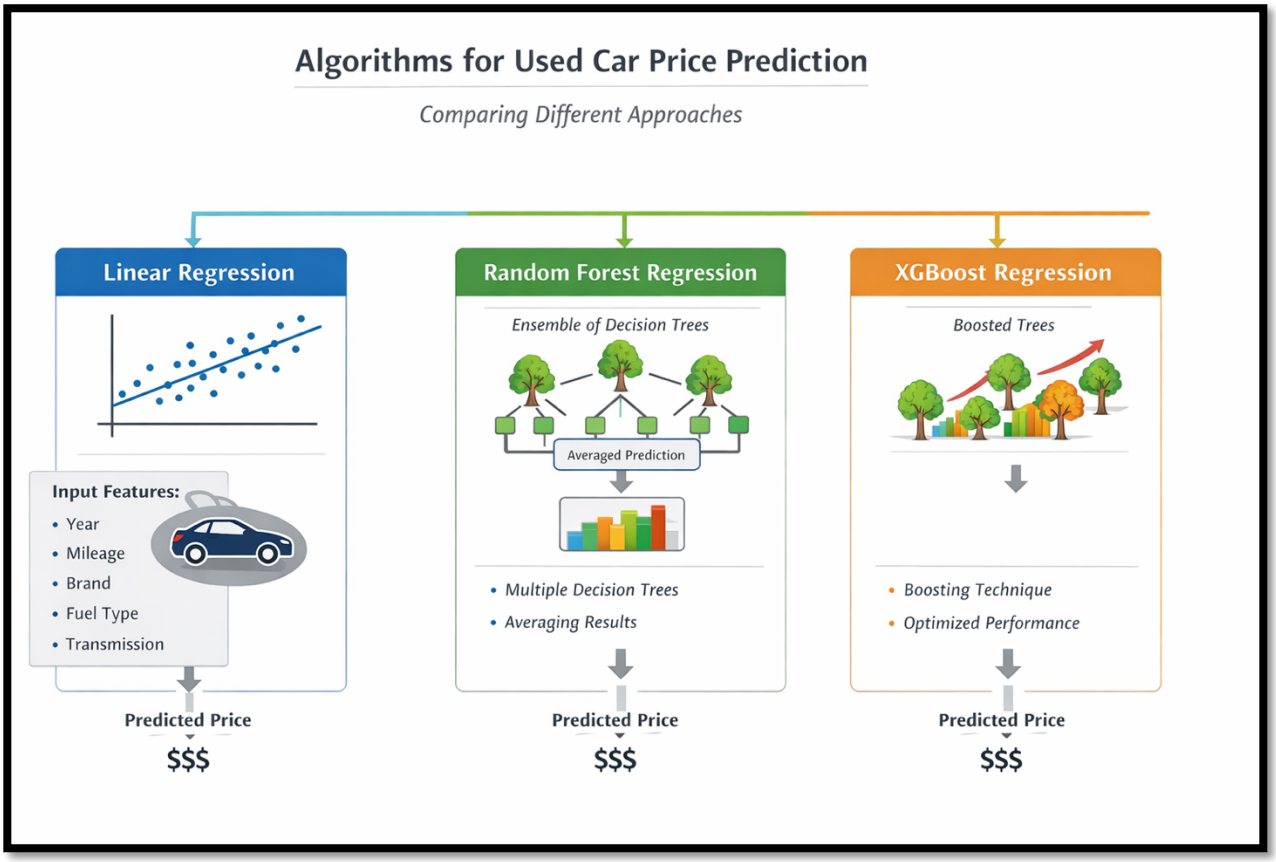


*Figure 7. Comparison among different algorithms*

The above image shows the comparison among three different algorithms to predict the price of used cars. Even though the end goal is the same for all models, the way of executing the process is vastly different where each algorithm has its own unique way of evaluating the target variable i.e. car price.

## 3.2 Diagrammatical Representations

To show the general workflow of all the algorithms, flowcharts and a state transition diagram is prepared and they are shown below:

### 3.2.1 Flowchart for Random Forest Regression

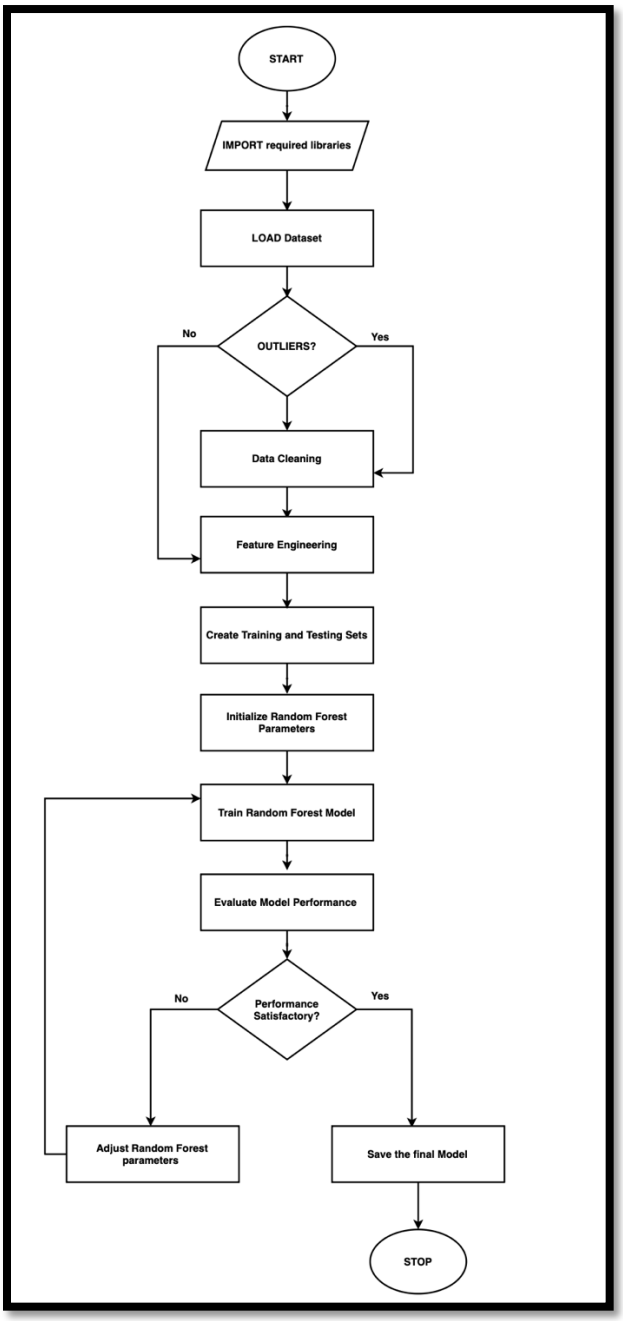


*Figure 8. Flowchart for Random Forest Regressor*

### 3.2.2 Flowchart for Linear Regression

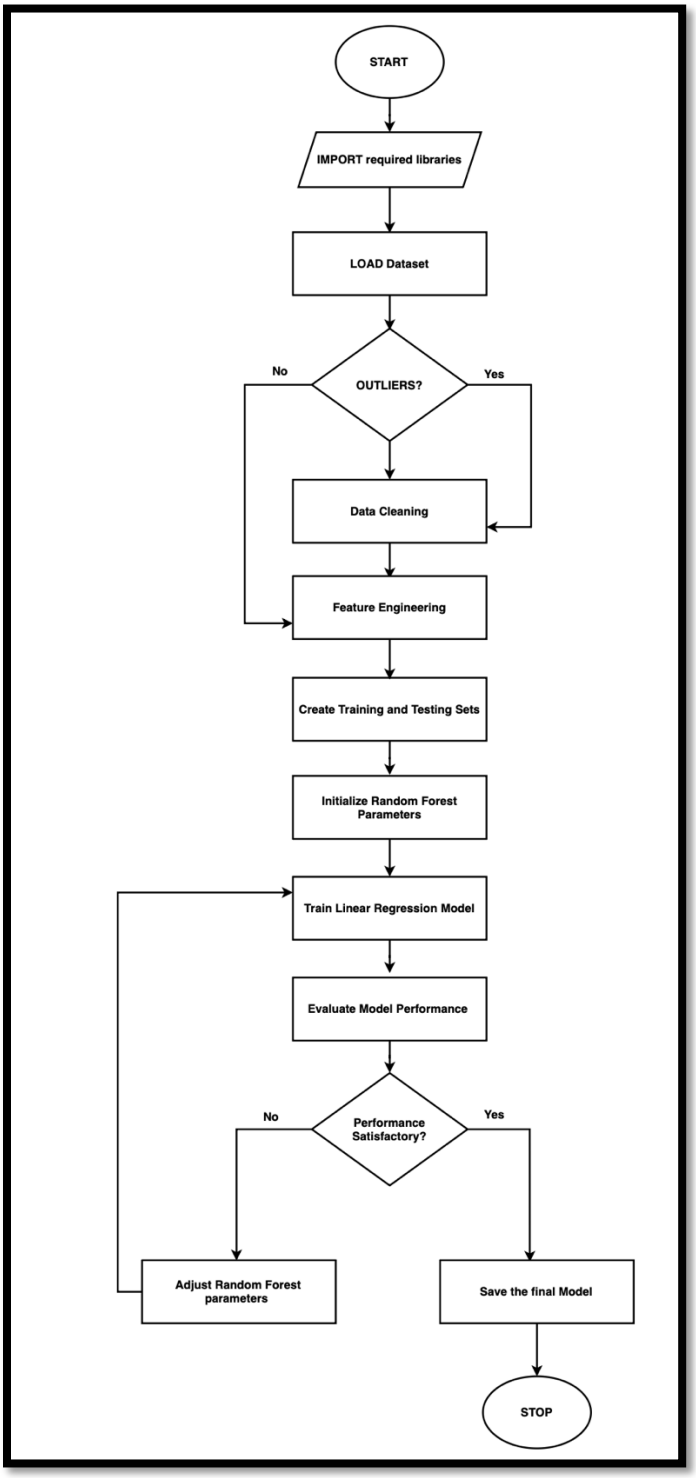


*Figure 9. Flowchart for Linear Regression*

### 3.2.3 Flowchart for XGBoost Regression

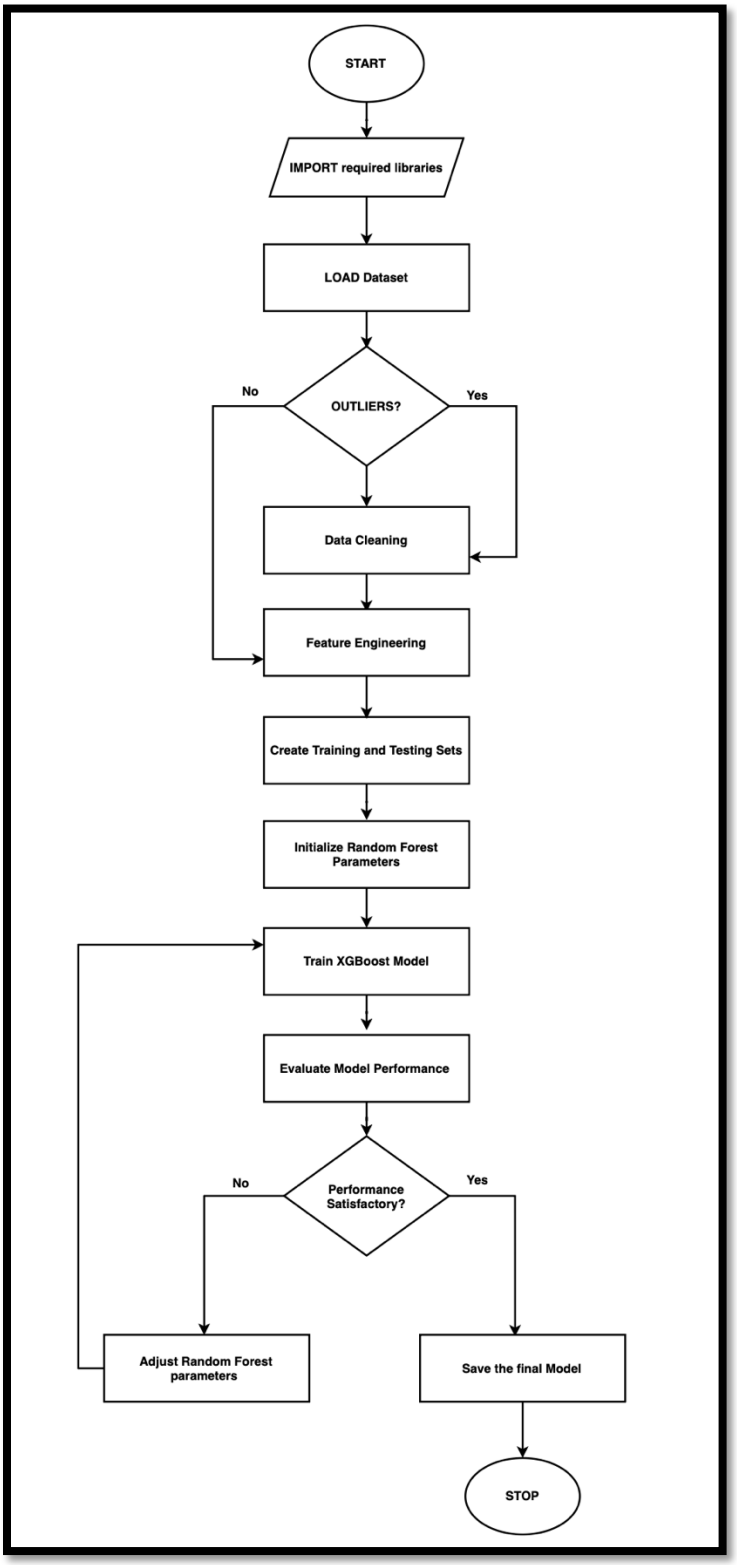


*Figure 10. Flowchart for XGBoost Regression*

### 3.2.4 State Transition Diagram

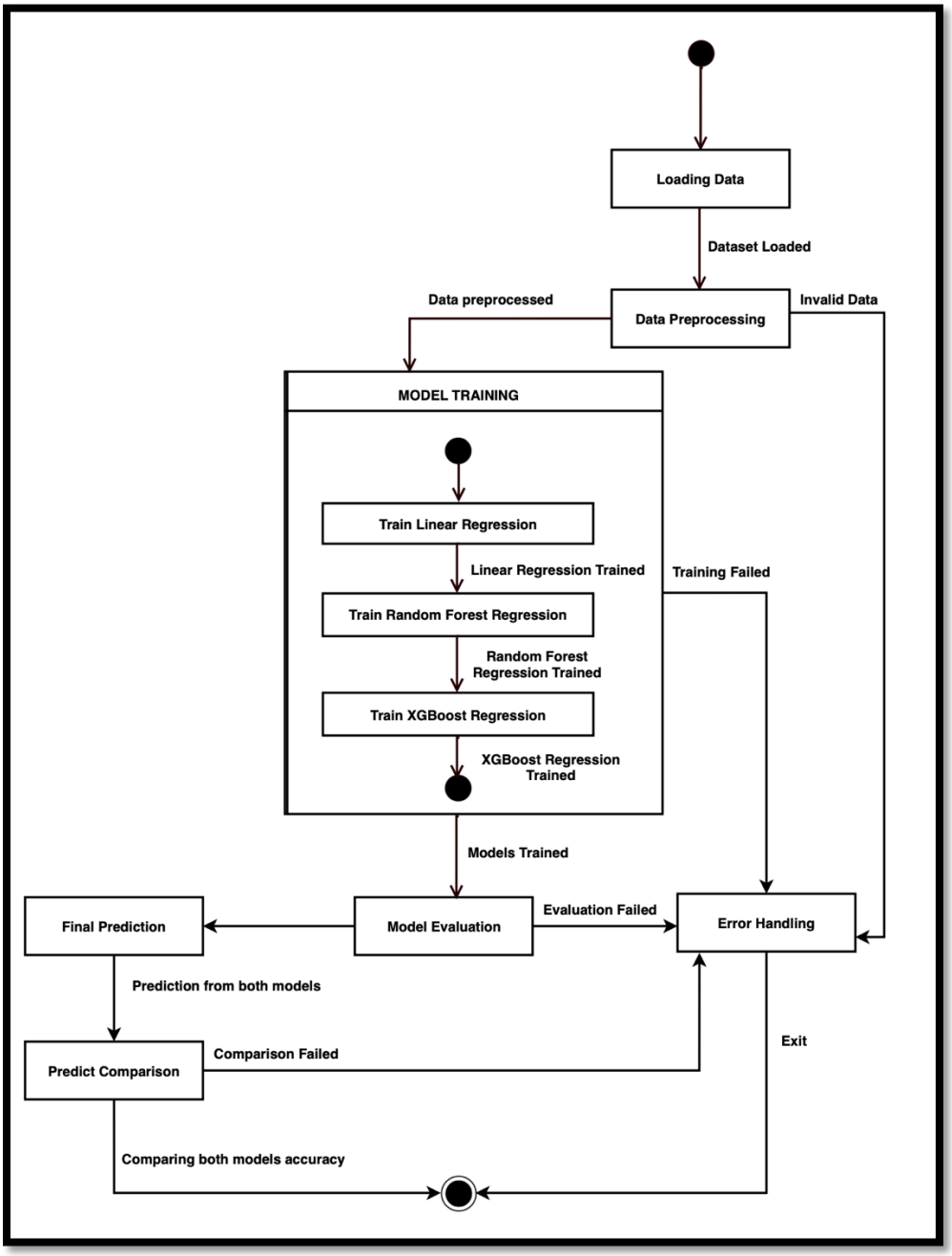


*Figure 11. State Transition Diagram*

## 3.3 Pseudocode of the Solution

### 3.3.1 Pseudocode for Random Forest Regression

**START**

// 1. IMPORT LIBRARIES

**IMPORT** pandas, numpy

**IMPORT** matplotlib, seaborn

**IMPORT** sklearn libraries (RandomForestRegressor, train_test_split, metrics)

**IMPORT** evaluation metrics (MAE, MSE, RMSE, R2)

// 2. LOAD DATASET

**LOAD** Car Price dataset from Hugging Face

**DISPLAY** dataset shape, columns, data types

// 3. DATA EXPLORATION

**EXPLORE** dataset structure

**CHECK** missing values in all columns

**CHECK** duplicate records

**ANALYZE** statistical summary of all features

**VISUALIZE** distribution of Price

**ANALYZE** correlation between features

**IDENTIFY** outliers in numerical features (Year, Kms_Driven, Mileage, Engine, Power)

```
// 4. DATA CLEANING
HANDLE missing values
    IF missing values exist
        REMOVE rows
    END IF
REMOVE duplicate records IF exist
HANDLE outliers in numerical features

// 5. FEATURE ENGINEERING
ENCODE categorical variables (Fuel_Type, Seller_Type, Transmission, Owner)
SCALE numerical features (optional for Random Forest)
SELECT relevant features
    ANALYZE feature importance
    REMOVE irrelevant features IF any
END SELECT

// 6. SPLIT DATASET
SET target variable = Price
SET features = all other columns
DIVIDE into Training (80%) and Test (20%) sets
```

```
// 7. MODEL TRAINING
INITIALIZE Random Forest Regressor
SET hyperparameters (n_estimators, max_depth, min_samples_split, etc.)
TRAIN model on Training Set
    BUILD multiple decision trees
    APPLY bootstrap sampling
    SELECT random feature subsets for each tree
END TRAIN

// 8. MODEL EVALUATION
PREDICT Price on Test Set
CALCULATE MAE, MSE, RMSE, R2
PLOT actual vs predicted prices
EXTRACT feature importance scores

// 9. MODEL OPTIMIZATION
WHILE performance not satisfactory
    TUNE hyperparameters
    RETRAIN model
    RE-EVALUATE performance
    ANALYZE feature importance changes
END WHILE
```

// 10. FINALIZATION

**SELECT** best model configuration

**SAVE** model

**DOCUMENT** performance metrics and important features

**STOP**

### 3.3.2 Pseudocode for Linear Regression

**START**

// 1. IMPORT LIBRARIES

**IMPORT** pandas, numpy

**IMPORT** matplotlib, seaborn

**IMPORT** sklearn libraries (LinearRegression, train_test_split, metrics)

**IMPORT** evaluation metrics (MAE, MSE, RMSE, R2)

// 2. LOAD DATASET

**LOAD** Car Price dataset from Hugging Face

**DISPLAY** dataset shape, columns, data types

// 3. DATA EXPLORATION

**EXPLORE** dataset structure

**CHECK** missing values in all columns

**CHECK** duplicate records

**ANALYZE** statistical summary of all features

**VISUALIZE** distribution of Price

**ANALYZE** correlation between features

**IDENTIFY** outliers in numerical features

// 4. DATA CLEANING

**HANDLE** missing values

  **IF** missing values exist

    REMOVE rows

  **END IF**

**REMOVE** duplicate records IF exist

**HANDLE** outliers in numerical features

// 5. FEATURE ENGINEERING

**ENCODE** categorical variables

**SCALE** numerical features (recommended for Linear Regression)

**SELECT** relevant features

  **ANALYZE** correlation with target

  **REMOVE** irrelevant features IF any

**END SELECT**

```
// 6. SPLIT DATASET
SET target variable = Price
SET features = all other columns
DIVIDE into Training (80%) and Test (20%) sets

// 7. MODEL TRAINING
INITIALIZE Linear Regression model
TRAIN model on Training Set
    CALCULATE coefficients using least squares method
END TRAIN

// 8. MODEL EVALUATION
PREDICT Price on Test Set
CALCULATE MAE, MSE, RMSE, R2
PLOT actual vs predicted prices

// 9. MODEL OPTIMIZATION
WHILE performance not satisfactory
    APPLY feature selection or transformation (log, polynomial)
    RETRAIN model
    RE-EVALUATE performance
END WHILE
```

// 10. FINALIZATION

**SAVE** model

**DOCUMENT** performance metrics and coefficients

**STOP**

### 3.3.3 Pseudocode for XGBoost Regression

**START**

// 1. IMPORT LIBRARIES

**IMPORT** pandas, numpy

**IMPORT** matplotlib, seaborn

**IMPORT** xgboost library (XGBRegressor)

**IMPORT** sklearn libraries (train_test_split, metrics)

**IMPORT** evaluation metrics (MAE, MSE, RMSE, R2)

// 2. LOAD DATASET

**LOAD** Car Price dataset from Hugging Face

**DISPLAY** dataset shape, columns, data types

// 3. DATA EXPLORATION

**EXPLORE** dataset structure

**CHECK** missing values in all columns

**CHECK** duplicate records

**ANALYZE** statistical summary of all features

```
VISUALIZE distribution of Price
ANALYZE correlation between features
IDENTIFY outliers in numerical features

// 4. DATA CLEANING
HANDLE missing values
    IF missing values exist
        REMOVE rows
    END IF
REMOVE duplicate records IF exist
HANDLE outliers in numerical features

// 5. FEATURE ENGINEERING
ENCODE categorical variables
SCALE numerical features
SELECT relevant features
    ANALYZE feature importance
    REMOVE irrelevant features IF any
END SELECT
```

// 6. SPLIT DATASET

**SET** target variable = Price

**SET** features = all other columns

**DIVIDE** into Training (80%) and Test (20%) sets

// 7. MODEL TRAINING

**INITIALIZE** XGBRegressor

**SET** hyperparameters (n_estimators, max_depth, learning_rate, subsample, colsample_bytree)

**TRAIN** model on Training Set

**BOOST** weak learners iteratively

**MINIMIZE** prediction error using gradient boosting

**END TRAIN**

// 8. MODEL EVALUATION

**PREDICT** Price on Test Set

**CALCULATE** MAE, MSE, RMSE, R2

**PLOT** actual vs predicted prices

**EXTRACT** feature importance scores

```
// 9. MODEL OPTIMIZATION
WHILE performance not satisfactory
    TUNE hyperparameters
    RETRAIN model
    RE-EVALUATE performance
    ANALYZE feature importance changes
END WHILE

// 10. FINALIZATION
SELECT best model configuration
SAVE model
DOCUMENT performance metrics and important features
STOP
```

## 4. Conclusion

This proposed project aims to develop a car price prediction system using three machine learning models which are Linear Regression, Random Forest Regression and XGBoost Regression. The study plans to analyse how different car attributes such as age, mileage, brand, fuel type, and engine specifications influence car prices. Linear Regression will be used as a baseline model to understand linear relationships, while Random Forest and XGBoost are proposed to capture non-linear patterns and improve prediction accuracy through ensemble learning. Comparative analysis of these models is intended to identify the most effective approach for car price prediction.

Coming up with second-hand vehicle prices is mostly based on personal opinions hence full of discrepancies, with very unfair valuations to both buyers and sellers. In these research works, the proposed solution is a data-driven, transparent pricing mechanism to assist an individual, dealer, or online marketplace in evaluating the fair price of a car. The system should reduce human bias, enhance decision-making, and improve trust in the used car market by using suitable machine-learning techniques that include predictive models. The system also solves a real-world problem as it also provides researchers with a trustworthy system which can be used for research purposes as well.

Further work includes setting the collection and preprocessing of real-world datasets, model training, and evaluation against suitable performance metrics for the selection of the best-performing model. An addition that could be considered is regional pricing trends, along with market demand, to increase accuracy. Another major further work could be country specific second-hand car prediction system where the dataset needs to be modified according to the country needs. For example, In the context of Nepal, used cars prices are also based on number of services pending or due of the car, insurance etc. This system could also be expanded into a web-based application meant for real-time prediction of the pricing of cars.

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